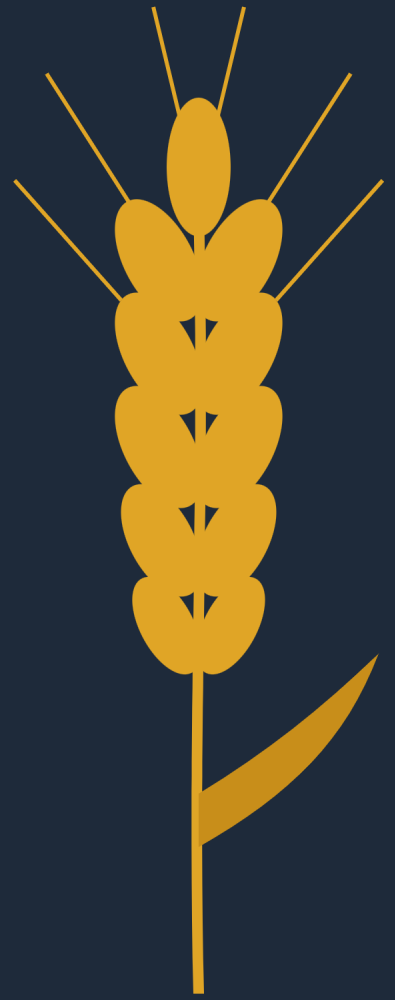
ICONIC CLUB AFACERI.RO · IAȘI

Financial Instruments

Explained with one bushel of wheat

Iulian Costan



OVERVIEW

What we'll cover today

PART ONE

Spot (cash) market

- **Spot** · own it today
- **Pure arbitrage** · market equilibrium

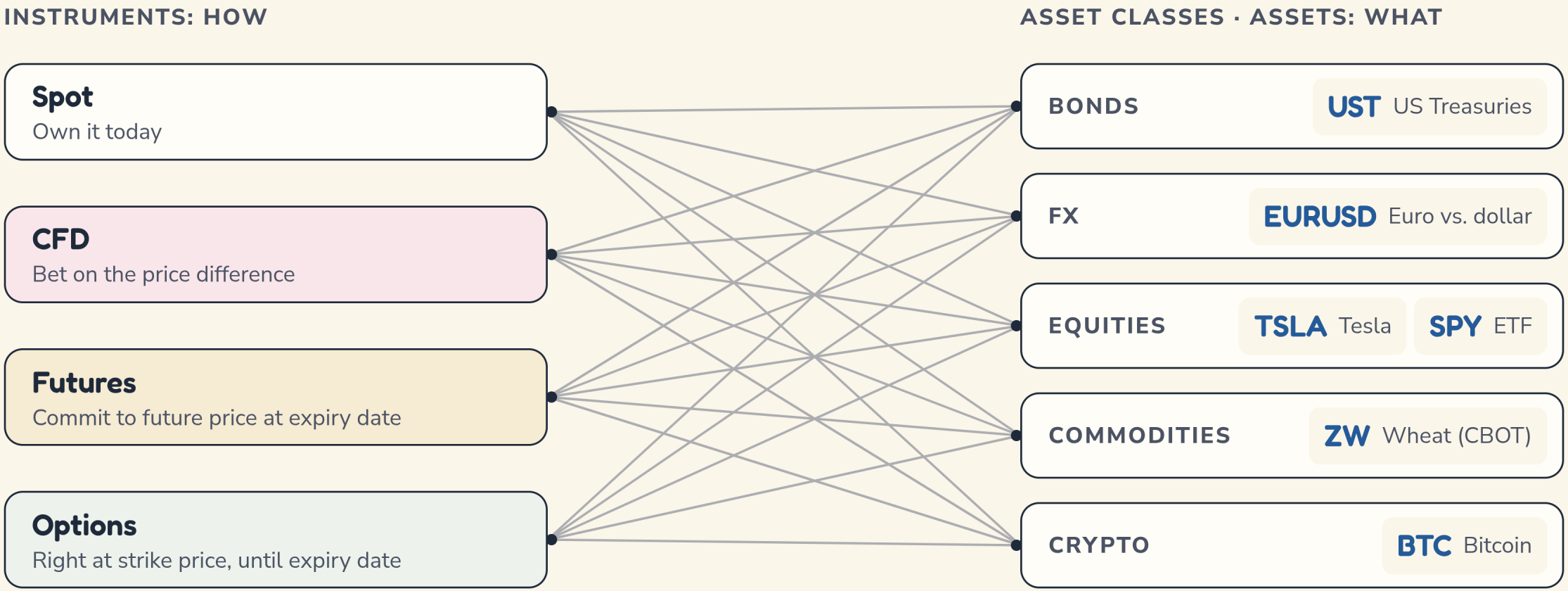
PART TWO

Derivatives

- **CFD** · leverage and margin
- **Futures** · farm, mill, basis
- **Options** · floor, ceiling, strike
- **Statistical arbitrage** · market equilibrium

OVERVIEW · ASSETS AND ASSET CLASSES

Any instrument, any asset



BEFORE WE START

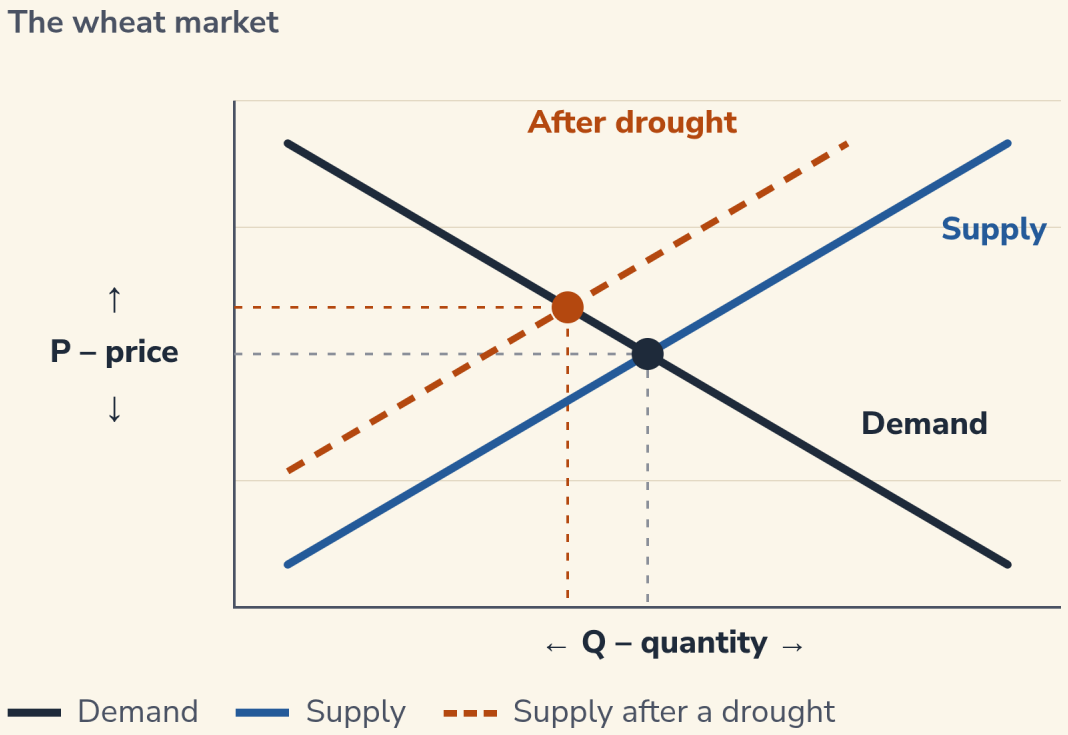
Where does a price come from?

Demand (mills, bakeries): they buy more when wheat is cheap.

Supply (farms): they sell more when wheat is expensive.

The price settles where the two meet: **$100 a bushel**.

⚠ **A drought cuts supply:** the price climbs to about $106.



PART ONE

Spot (cash) market

You pay the full price, and the thing is yours, today.

BID $99

The best price a buyer will **pay you**. You sell at the bid.

SPREAD $2

Ask minus bid: the **cost** of trading right now.

ASK $101

The best price a seller will **take from you**. You buy at the ask.

Wheat at the silo, or stocks, bonds and ETFs on the exchange.

1 · SPOT MARKET

Pay now, get it now

At the silo in Chicago

You pay $100 and get a bushel of wheat. On the spot. It's yours, and you take it home.

On the exchange

You pay $100 and get stocks, bonds or ETFs. They land in your account, and they're yours.

Spot means you own it. Settlement on an exchange takes a day or two, but it's still "now".

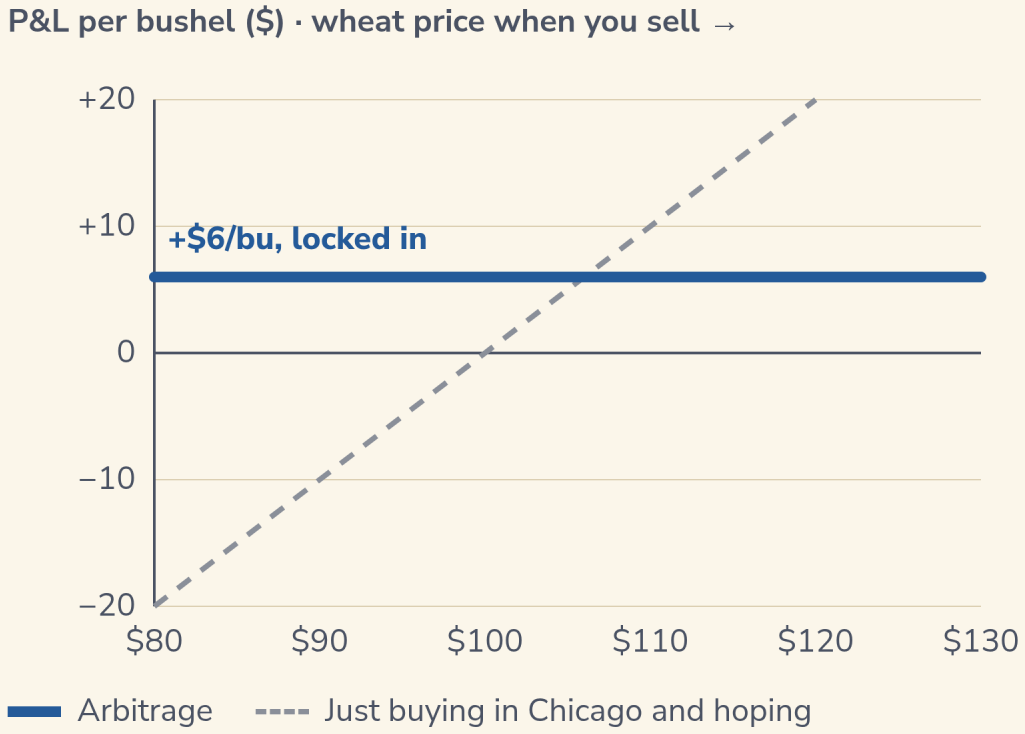
1 · PURE ARBITRAGE

Same wheat, two prices

Per bushel	$
Buy in Chicago	100
+ Transport	4
+ Kiosk and fees	2
Sell in Detroit	112

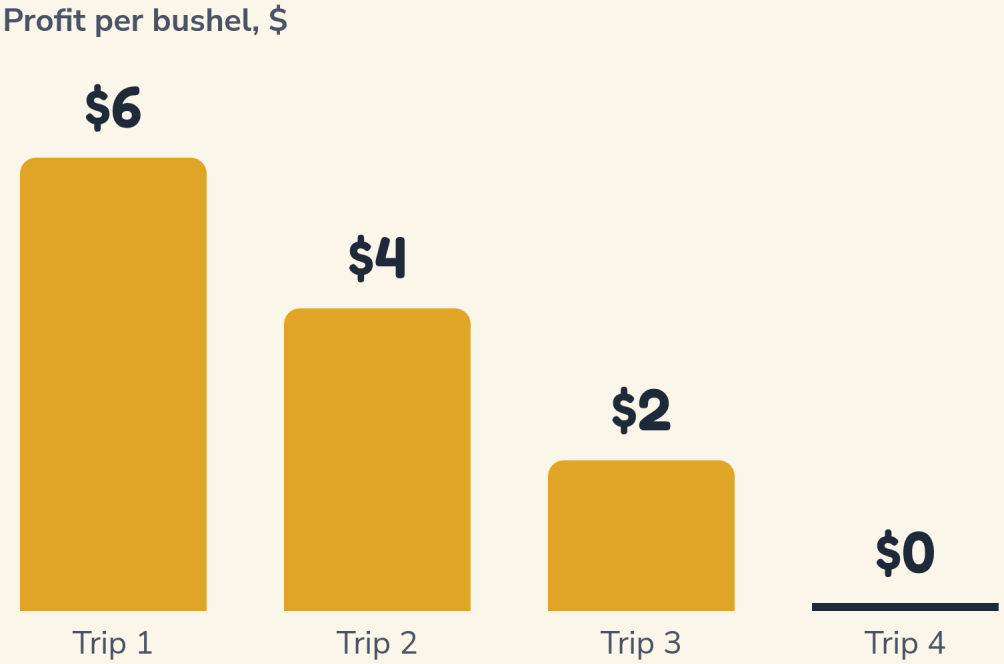
Profit: $6 per bushel

Buy and sell at the same moment. No forecast needed.



1 · PURE ARBITRAGE

Every truckload shrinks the edge



We buy in Chicago, so demand there grows and the price **rises**. We sell in Detroit, so supply there grows and the price **falls**.

The gap ends up covering only transport and the kiosk. The market is back in **equilibrium**.

And someone faster usually spots the gap first.

PART TWO

Derivatives

You don't buy the wheat. You buy a contract on the price of wheat.

LONG ↑

You **buy** the contract. You profit when the price **rises**.

SHORT ↓

You **sell** the contract first. You profit when the price **falls**.

The contract's value "derives" from something else: wheat, oil, stocks, indices, currencies.

2 · LEVERAGE AND MARGIN

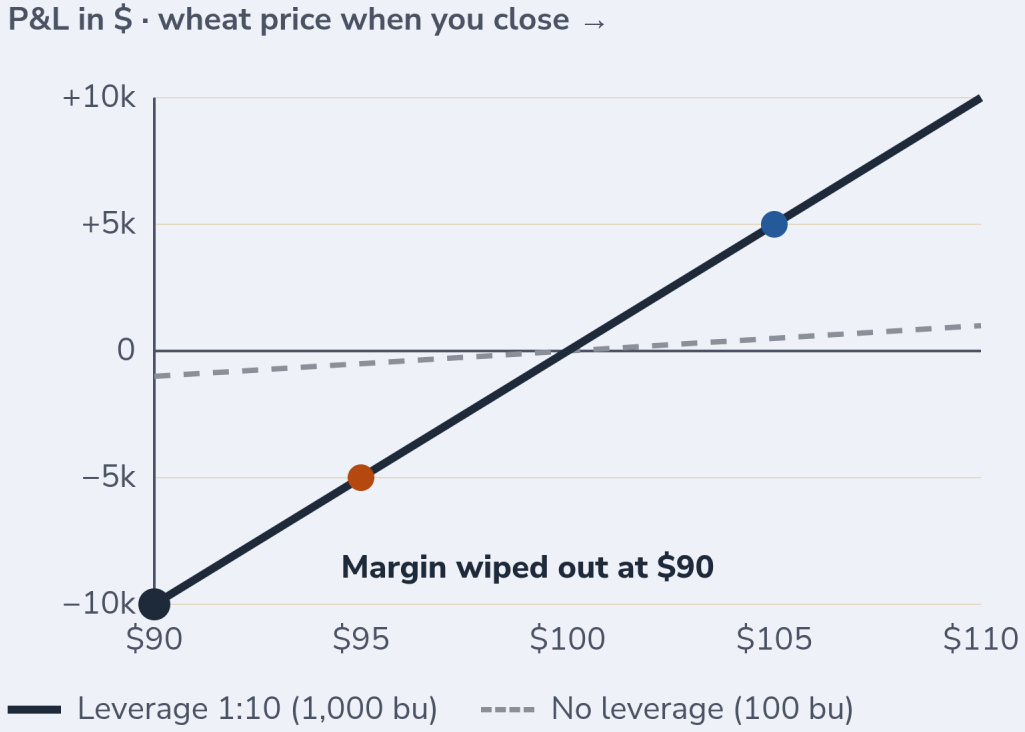
$10,000 that acts like $100,000

Margin $10,000 × leverage 1:10 = exposure $100,000, i.e. 1,000 bushels at $100.

WHEAT +5%
Gain: +$5,000
+50%

WHEAT −5%
Loss: −$5,000
−50%

WHEAT −10%
Margin call: position closed
$0



2.1 · CFD

CFD: a bet on the price difference

Contract for Difference. You and the broker settle the gap between entry and exit price.

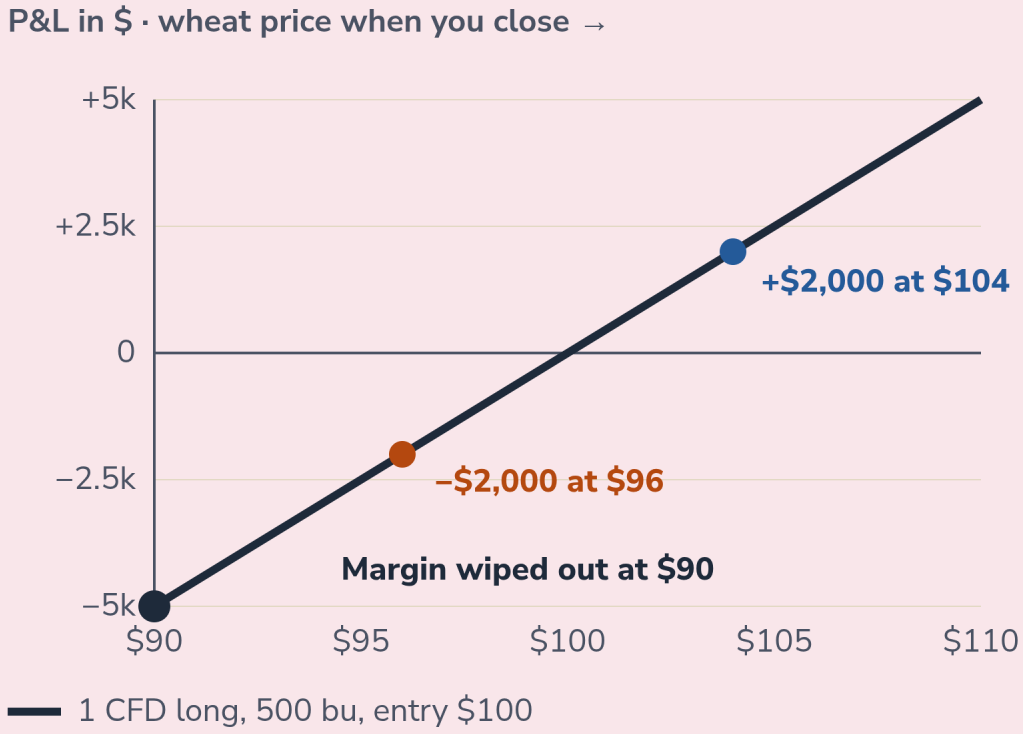
Example: wheat CFD

Contract size: **500 bushels**

At $100/bu → exposure **$50,000**

10% margin: **$5,000**

Unlike spot: you never own the wheat, you pay financing every night, and the broker is your counterparty.



2.2 · FUTURES – INTRODUCTION

Futures: commit to future price at expiry date

FUTURE PRICE

$105

The price agreed **today** for wheat delivered later. Buyer and seller are both **committed** to it, whatever spot does.

EXPIRY DATE

September

The day the contract **ends**: the wheat is delivered or the difference is paid, and futures meet spot.

2.2 · FUTURES – FARM EXAMPLE

The farm locks in the price in March

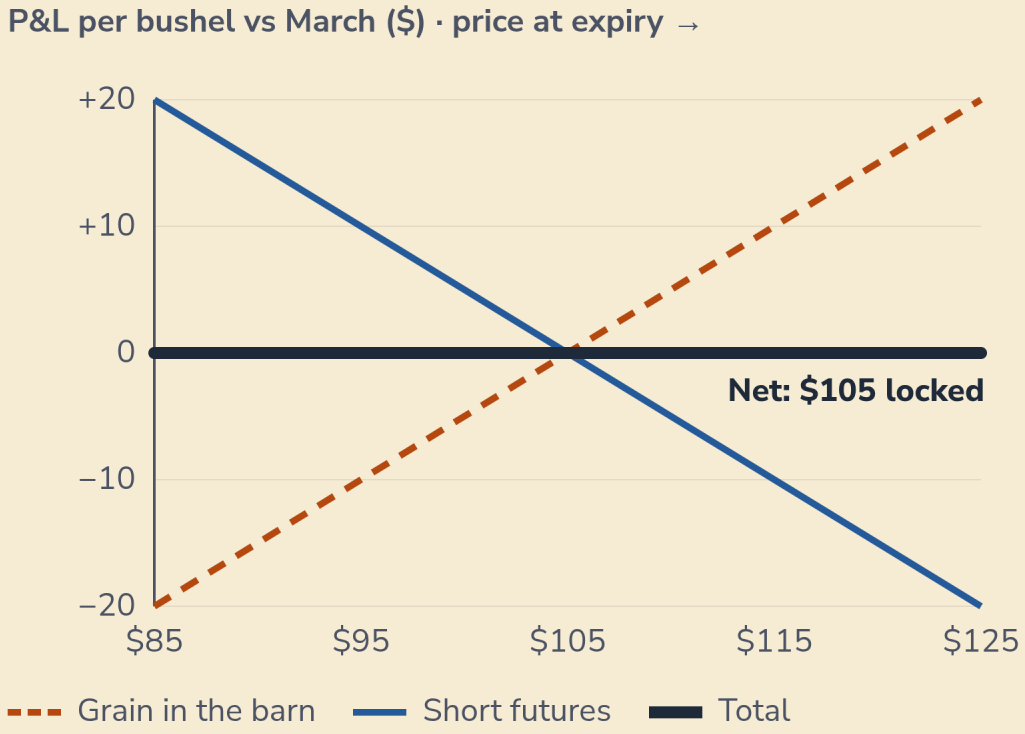
March spot: **$110**

Sep futures: **$105**

Its crop is still in the field, so it can't sell spot. It **sells (goes short) September futures at $105.**

September	Grain sale	Futures	Net /bu
Falls to $90	$90	+$15	$105
Rises to $120	$120	–$15	$105

$105/bu either way. Easier credit, better planning.



2.2 · FUTURES – MILL EXAMPLE

The mill locks in the cost of flour

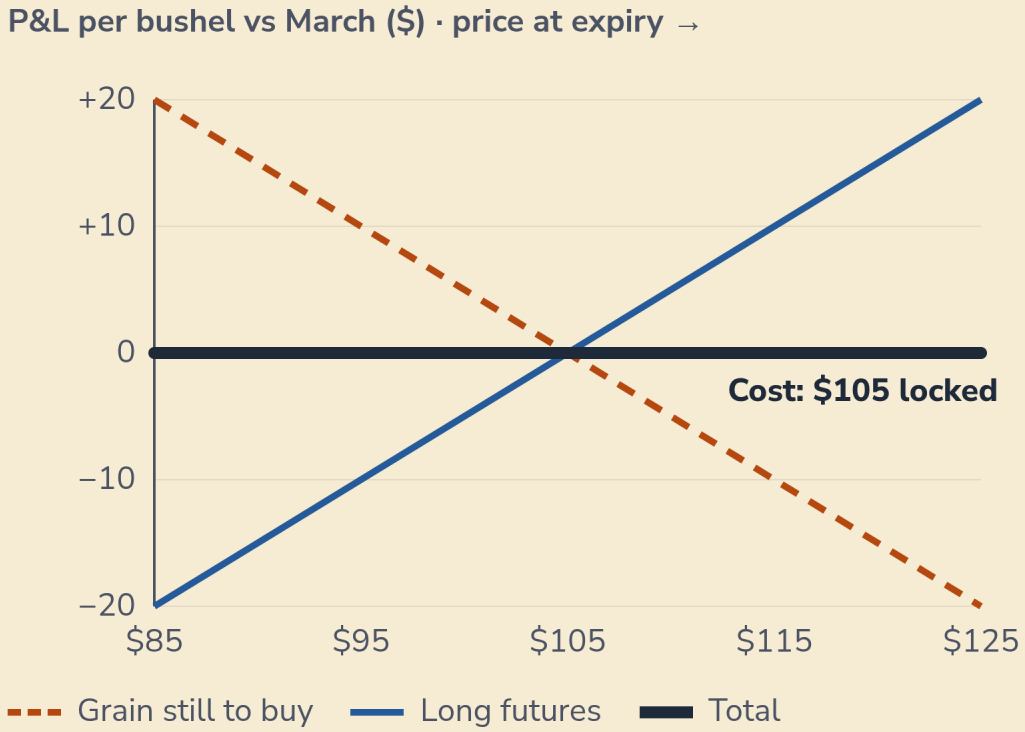
March spot: **$110**

Sep futures: **$105**

It needs wheat in September, not now, and fears a drought. It **buys (goes long) September futures at $105.**

September	Grain cost	Futures	Net /bu
Rises to $120	$120	+$15	$105
Falls to $90	$90	–$15	$105

A fixed cost: it can sign year-long contracts with bakeries.



2.2 · FUTURES – BASIS

Basis: spot minus futures

July: harvest

Wheat everywhere, silos are full, storage is expensive.

Spot **$90** · December futures **$98**

Basis –$8: you pay to wait

What it means: futures sit above spot by the cost of storage. The market pays whoever stores wheat until December.

March: before harvest

Silos are almost empty and the mills are hunting for wheat.

Spot **$110** · September futures **$105**

Basis +$5: wheat today is scarce

What it means: spot sits above futures because wheat is short now. The market pays whoever sells from the silo today.

At expiry the basis goes to zero: futures and spot meet.

2.3 · OPTIONS – INTRODUCTION

Options: a right at strike price, until expiry date

CALL

The right to **buy** at a price fixed today. It puts a ceiling on your cost.

PUT

The right to **sell** at a price fixed today. It puts a floor under your price.

You pay a **premium**, like insurance. If you don't need it, you let it expire and lose only the premium.

2.3 · OPTIONS – FARM EXAMPLE

The farm buys a floor

March spot: **$110**

Sep futures: **$105**

Crop still in the field: no spot sale in March. **$105 in September is fine; below $95 the farm is in trouble.**

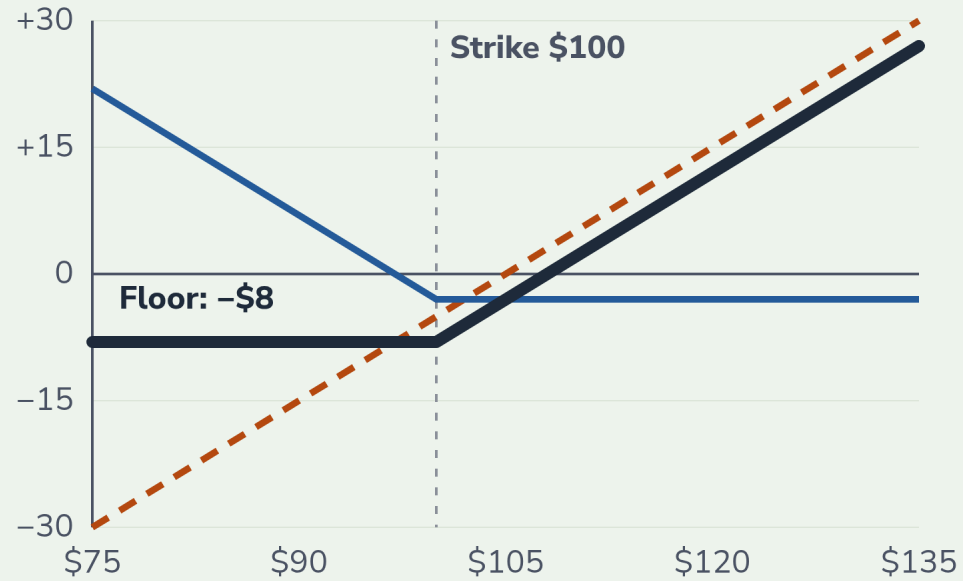
Buys a PUT · strike $100 · premium $3/bu

September	Grain sale	Put	Net /bu
Falls to $80	$80	+$17	$97
Falls to $90	$90	+$7	$97
Rises to $120	$120	–$3	$117

Put column after the $3 premium: +$10 payout – $3 = +$7.

Floor at $97, and the upside stays its own.

P&L per bushel vs Sep futures $105 · price at expiry →



–$8 = strike $5 below futures ($100 vs $105) + $3 premium

2.3 · OPTIONS – MILL EXAMPLE

The mill buys a ceiling

March spot: **$110**

Sep futures: **$105**

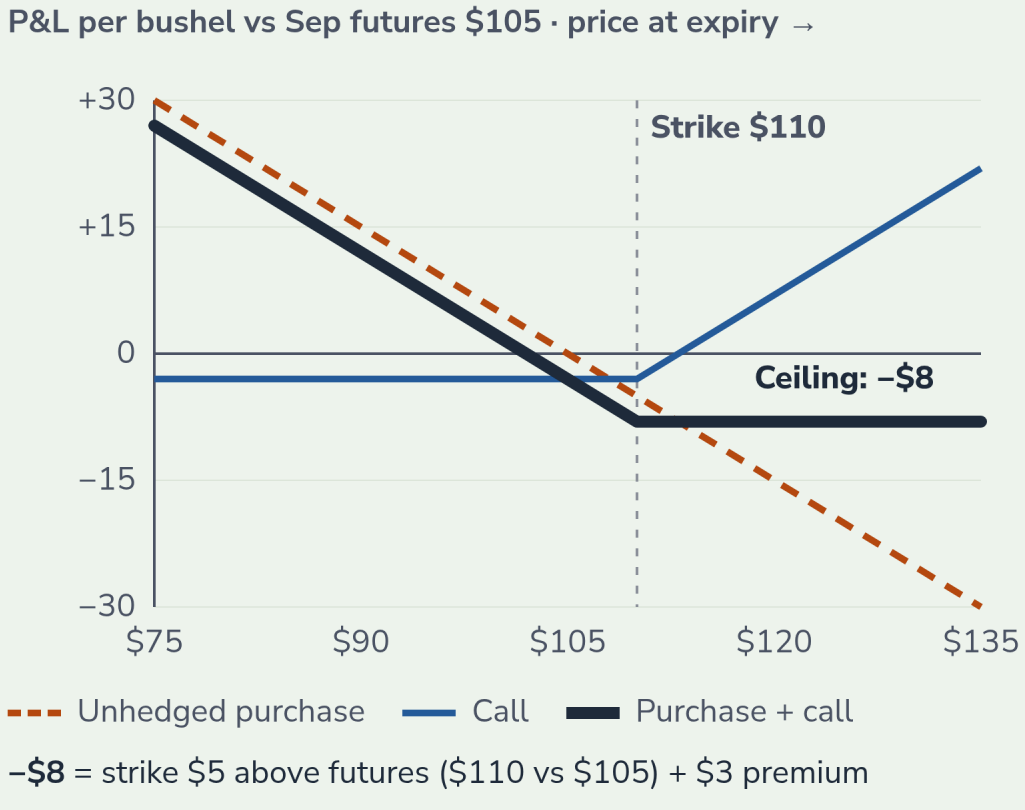
It needs wheat in September, not now. **$105 is fine; above $115 the business is in danger.**

Buys a CALL · strike $110 · premium $3/bu

September	Grain cost	Call	Net /bu
Rises to $130	$130	+$17	$113
Rises to $120	$120	+$7	$113
Falls to $90	$90	–$3	$93

Call column after the $3 premium: +$10 payout – $3 = +$7.

Ceiling at $113, and cheaper wheat stays its gain.



–$8 = strike $5 above futures ($110 vs $105) + $3 premium

2.3 · OPTIONS – STRIKE PRICE

Strike: where the option starts to pay

Wheat at expiry	Farm's PUT, strike $100	Mill's CALL, strike $110
$90 (big crop)	In the money: worth $10	Out of the money: $0
$105 (normal year)	Out of the money: $0	Out of the money: $0
$120 (drought)	Out of the money: $0	In the money: worth $10

STRIKE

The fixed price written in the option: the price you may sell at (put) or buy at (call).

SPOT VS STRIKE

Decides the value at expiry: in the money it pays, out of the money it's worth $0.

LIKE BASIS FOR FUTURES

A better strike (higher floor, lower ceiling) costs a higher premium.

You win on average over a large number of trades

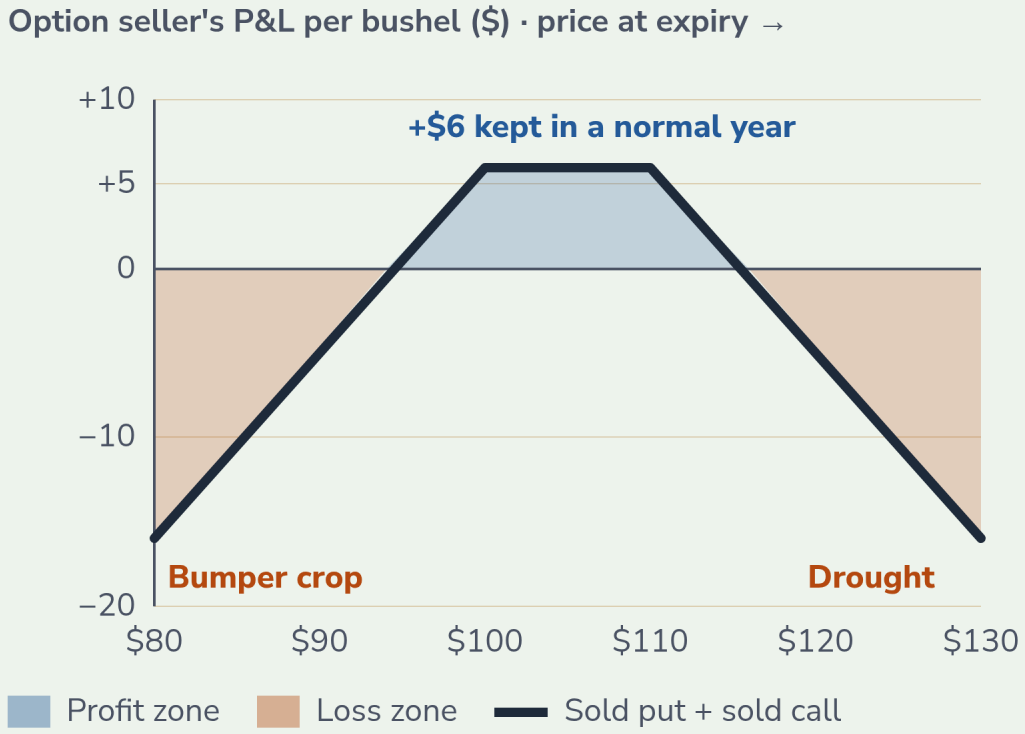
PURE ARBITRAGE

High edge (high expected value, high probability of profit) per trade.

STATISTICAL ARBITRAGE

A small edge, repeated thousands of times.

Who sells the farm's put and the mill's call? A trader acting like an **insurer**: keeps $6/bu in a normal year, pays out in extreme ones.



Three things to remember

1. Spot means you own it. A derivative is a contract on the price.
2. Leverage magnifies everything: gains and mistakes.
3. Hedging isn't a bet. It's insurance for the business.

Questions?

Educational material, not investment advice. Prices in the examples are illustrative.